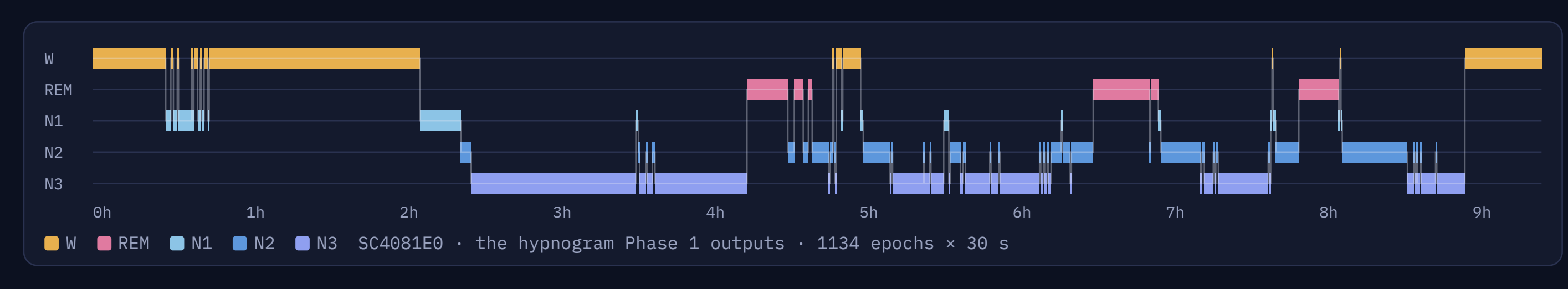
SLEEP-STAGING · TEAM 40 · PIPELINE MAP · 20 OCT 2026

From one night of EEG to a verified sleep report

What goes into each phase, what happens to it, and what comes out. Every number and example on this page comes from the repository, mostly from dev night SC4081E0.

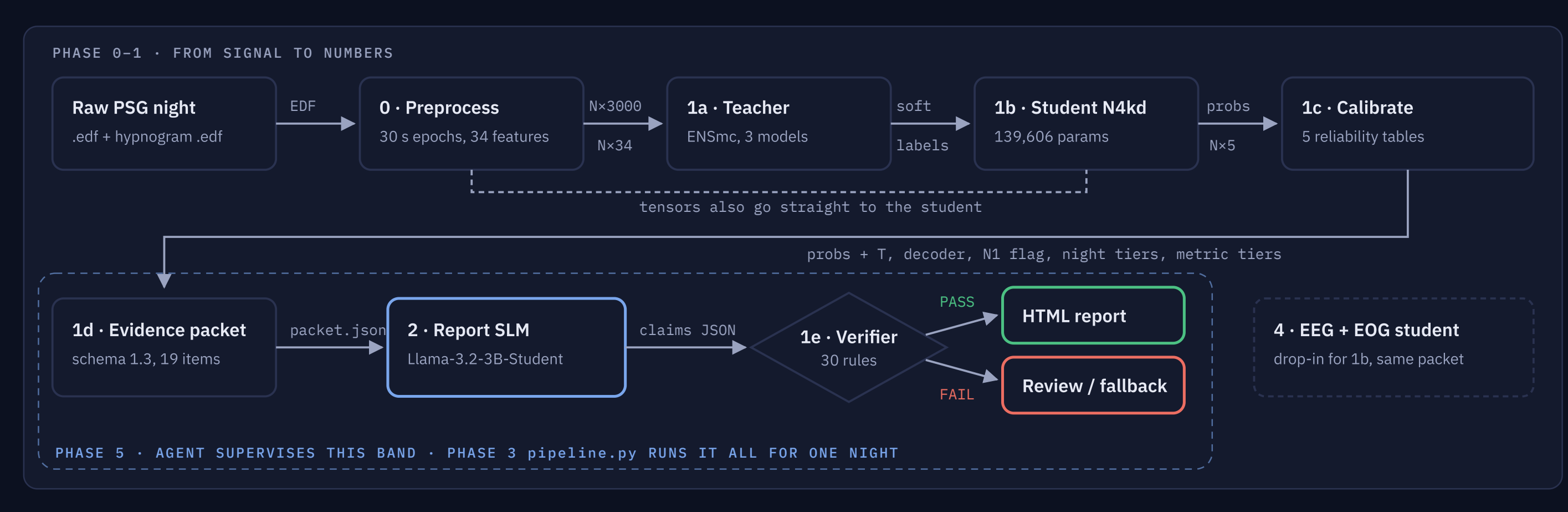


SC4081E0 · the hypnogram Phase 1 outputs · 1134 epochs × 30 s

THE WHOLE FLOW

Signal → numbers → packet → claims → gate → report

Each box is a phase; the text between boxes is what it hands to the next one. Click a box to jump to its details.



THE DATA

197 recordings, split by subject

Train 137	Dev 31	Test 29 · locked
trains staging models; 123 + 14 train the report model	fits every table	read once, at the end

Sleep-EDF Expanded: SC (healthy) and ST (temazepam study) cohorts. No person appears in two splits. The test packets' checksum `050fffe4…dd92` has not changed since they were locked.

Phase 0 Preprocessing

INPUT
- `SC4081E0-PSG.edf` : the polysomnogram, 100 Hz
- `SC4081EC-Hypnogram.edf` : the expert's stage annotations

→

PROCESS
1. Read with MNE, pick EEG Fpz-Cz
2. Trim to the annotated study period
3. Cut into 30 s epochs of 3000 samples
4. Map annotations to W / N1 / N2 / N3 / REM
5. 34 features per epoch: 24 wavelet (db4) + 7 STFT band powers, entropy, edge frequency + 3 band ratios

→

OUTPUT
- `tensors/<id>.pt` : **[N × 3000]**
- `spectral/<id>_spectral.pt` : **[N × 34]**
- `index.csv` : one row per night with its stage sequence

`e.g. SC4001E0 → [841 × 3000] ≈ 7 h` `code/Phase1_48/Phase1reworked/preprocess.py` `distillation/preprocess_multichannel.py`

> Found 11 Oct: trimming removed unscored epochs from the labels but not the signal, so 5–7 % of labels are shifted. Corrected labels are in `ADDING EOG VERSION - ST NIGHTS/results/`.

Phase 1a · 1b Teacher, then a smaller student

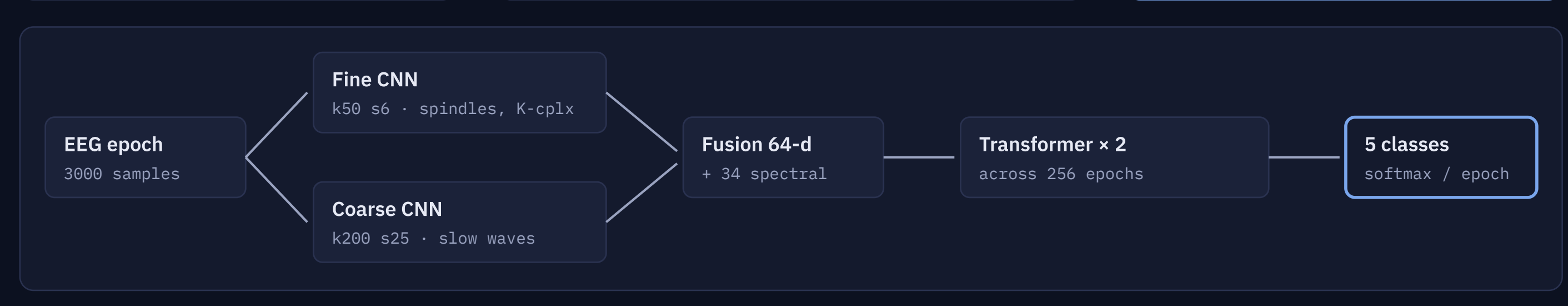
INPUT
- Train tensors + spectral features + labels
- For the student: the teacher's per-epoch probabilities (soft labels)

→

PROCESS
1. **Teacher ENSmc**: average the softmax of 3 EEG models (412,311 params total)
2. **Student N4kd**: reads 256 epochs (~2 h) at a time
3. Loss = 0.5 × KD (T = 3) + 0.5 × class-weighted cross-entropy, 75 epochs

→

OUTPUT
- `student_N4kd/student_best.pt` , 139,606 params
- Per-epoch probabilities **[N × 5]** for W, N1, N2, N3, REM



TEST SET, 29 NIGHTS	ACCURACY	COHEN'S κ	PARAMS
Teacher ENSmc	78.68 %	0.7140	412,311
Student N4kd, as shipped (decoded)	77.69 %	0.6993	139,606
Student N4kd, corrected labels	79.00 %	0.7169	139,606

`distillation/build_ensemble.py` `distillation/cache_teacher_logits.py` `distillation/student_model.py` `distillation/kaggle_train_student_kd.py`

Phase 1c Calibration and reliability tables

Fitted on the 31 validation nights only. They turn raw probabilities into numbers whose error is known before any report is written.

TABLE	QUESTION IT ANSWERS	N4KD VALUE	FILE
Temperature	How to make confidence match accuracy	`T = 1.2847`	`reliability_table.json`
Sequence decoder	How to remove stray single REM epochs	`REM min-run 3`	`sequence_decoding.json`
N1 flag	Which N1 calls are low-confidence	`max p < 0.575`	`n1_flag.json`
Night confidence	Is this whole night easy or hard	`high ≤ 0.608 nats`	`night_confidence.json`
Metric reliability	Which clinic numbers can be stated plainly	`3 robust · 12 fragile · 9 unreliable`	`derived_metric_reliability.json`

Phase 1d The evidence packet

INPUT
- The night's probabilities from 1b
- The five tables from 1c

→

PROCESS
1. Calibrate, then decode the hypnogram
2. Compute clinic metrics: TST, efficiency, latencies, stage fractions
3. Attach each metric's validation error and tier
4. Set the night tier, flag weak N1 epochs, add attribution
5. List the 19 evidence items a report may cite

→

OUTPUT
- `phase2_dev_packets/SC4081E0-PSG.json`
- Schema 1.3 · 137 train / 31 dev / 29 test

The 19 evidence items for SC4081E0

`5 mandatory: must be stated as fact` `14 hedged: must carry error and caveat`

Total sleep time	**410 min**	`arch.total_sleep_time`
Time in bed	**537 min**	`arch.time_in_bed`
Sleep efficiency	**76.3 %**	`arch.sleep_efficiency`
Night confidence	**high tier**	`night.confidence`
N1 reliability	**low tier**	`model.n1_reliability_warning`
Sleep onset latency	**28.5 min**	`arch.sleep_onset_latency`
Wake after sleep onset	**98.5 min**	`arch.waso`
REM latency	**227.5 min**	`arch.rem_latency`
REM latency, sustained	**227.5 min**	`arch.rem_latency_sustained`
REM periods	**6**	`arch.rem_periods`
Stage transitions	**99**	`arch.stage_transitions`
Transition rate	**10.48 /h**	`arch.transition_rate`
Light / deep ratio	**0.713**	`arch.light_deep_ratio`
Wake interruptions	**1.76 /h**	`arch.wake_interruptions_per_hour`
Wake fraction	**27.7 %**	`stage.W.fraction`
N1 fraction	**6.3 %**	`stage.N1.fraction`
N2 fraction	**19.2 %**	`stage.N2.fraction`
N3 fraction	**35.9 %**	`stage.N3.fraction`
REM fraction	**10.9 %**	`stage.REM.fraction`

`distillation/build_packet.py`

Phase 2 The report model: packet in, claims out

INPUT
- One packet
- Prompt P1
- Grammar `report/claims.gbnf`

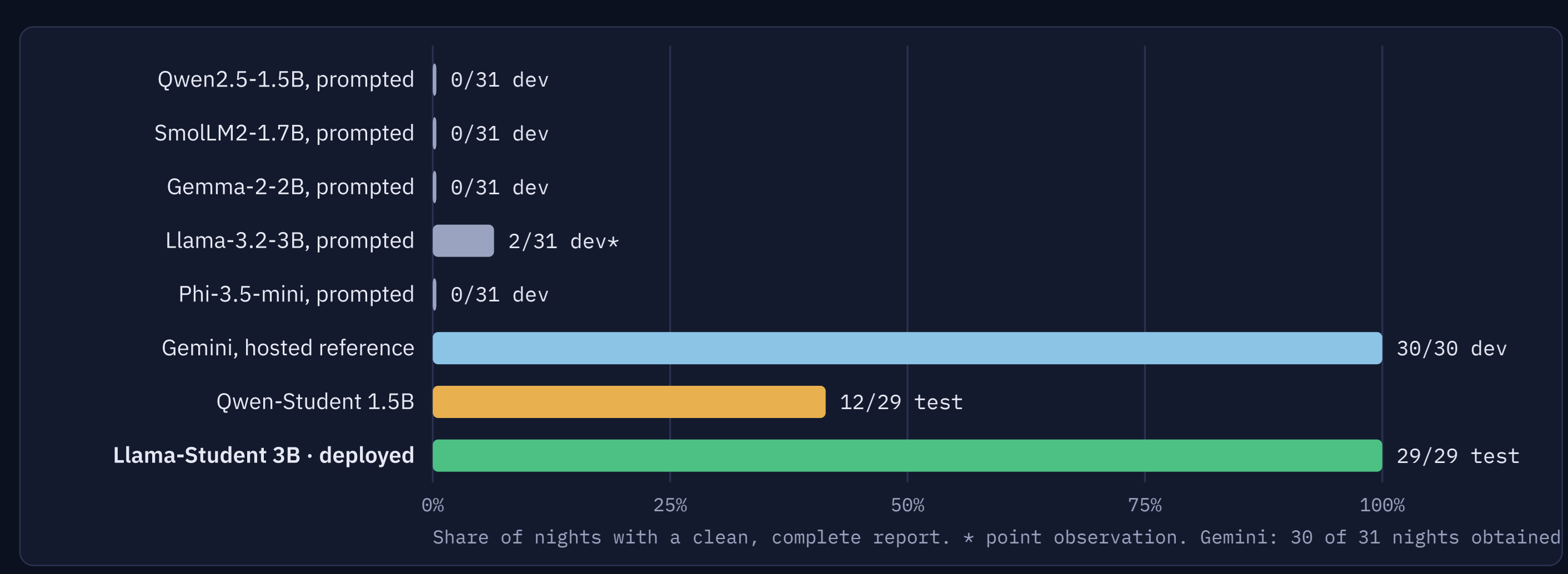
→

PROCESS
1. Llama-3.2-3B-Student (Q4_K_M, 2.0 GB) on CPU
2. Temperature 0, seed 0, one generation per night
3. The grammar makes malformed claims impossible to write

→

OUTPUT
- A JSON array of claims, each citing evidence IDs
- No prose: the model never writes the final English

How the model was chosen



Share of nights with a clean, complete report. * point observation. Gemini: 30 of 31 nights obtained

Untrained small models got the numbers right (one wrong number in 155 reports) but broke the rules: which items to hedge and which evidence to cite. The hosted reference model proved the contract can be met, so the pre-registered rule said train. Both students learned from 123 + 14 oracle examples (LoRA rank 16, 3 epochs, Kaggle T4). Llama won on the locked test set.

Phase 1e The verifier gate and the renderer

Frozen code in `report/`. A report renders only when every claim passes. The verifier never infers missing evidence: if the packet doesn't say it, the claim fails.

1 · CLAIM FROM THE MODEL

```
{"claim_id": "c2",
 "claim_type": "hedged_value",
 "cites": ["arch.rem_latency"],
 "subject": "this_recording",
 "value": 227.5,
 "unit": "minutes"}
```

2 · VERIFIER, 30 RULES

- ✓ Layer 1: valid JSON, known fields, unique ID
- ✓ Cited item exists in the packet
- ✓ 227.5 equals the packet value exactly
- ✓ Item is unsafe, so it must be hedged: it is
- ✓ All 5 mandatory items present in the set

3 · RENDERED FROM A FIXED TEMPLATE

REM latency: 227.5 minutes, mean absolute error 48.2 minutes on the validation split. Caveat: depends on a single epoch.

`deploy/reports/SC4081E0-PSG_2026-10-05.html`

`report/verify_structure.py` `report/verify_policy.py` `report/render.py`

Phase 3 Deployment: one new night, end to end

INPUT
- One raw `.edf`
- Hypnogram, if present, to find the study period

→

PROCESS · PIPELINE.PY
1. Preprocess → tensor
2. N4kd inference → probabilities
3. Build packet
4. Local Llama → claims → verifier

→

OUTPUT
- PASS: `deploy/reports/<id>_<date>.html`
- FAIL: `deploy/human_review/`
- Every run: `audit_log.jsonl`

`pinned by deploy/frozen.json` `~115 s per report on CPU` `4.7 GiB peak` `python pipeline.py --edf <night.edf>`

> Running the model through `llama-server` (what `pipeline.py` does) scores 20–23/31 on dev against 30/31 with the frozen `llama-completion` path. The verifier still blocks every bad report, but this should be fixed before the Raspberry Pi deployment.

Phase 4 EEG + EOG upgrade

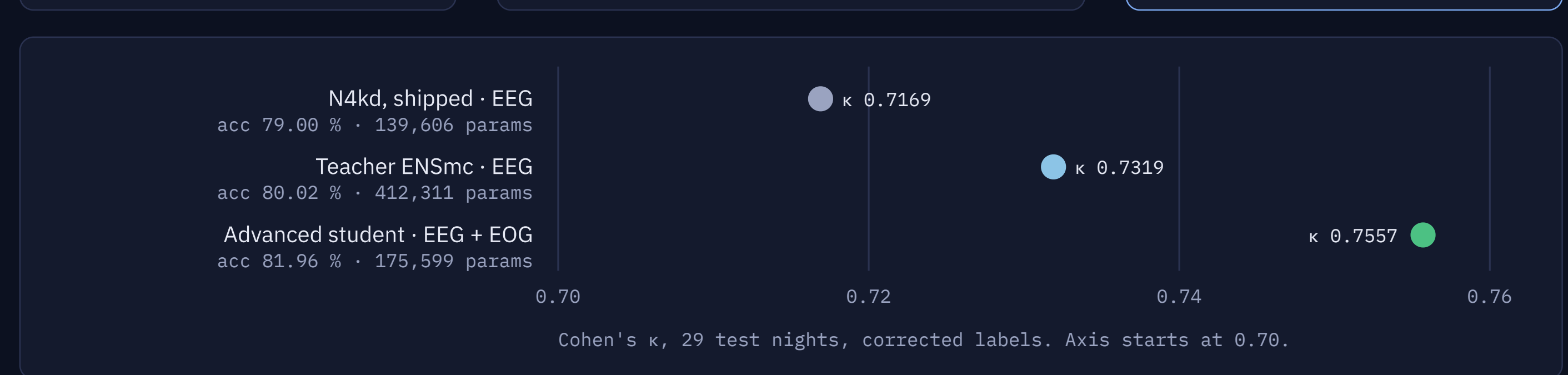
INPUT
- EEG + horizontal EOG
- Corrected labels

→

PROCESS
1. N4kd backbone plus gated EOG, time-frequency and local branches
2. Gates start closed, so training begins at N4kd
3. Refit all Phase 1c tables on validation, rebuild packets

→

OUTPUT
- `student_N6adv_full_s1` , 175,599 params
- Same packet schema: the frozen report model scores dev 30/31, no retraining



Cohen's κ, 29 test nights, corrected labels. Axis starts at 0.70.

Phase 5 The agent around the pipeline

The agent decides what to do: check, retry, fall back, deliver, audit. It never decides what is true. Stages come from the staging model, numbers from the packet, and every claim passes the frozen verifier.

INPUT
- Path A: a packet
- Path B: a raw EDF, no hypnogram

→

PROCESS · STATE MACHINE
1. Ingest, quality check, study period
2. Stage, confidence, packet
3. Wait for resources, then report and verify
4. Render, then V-HTML: every number on the page must trace to the packet
5. One retry on crash or timeout only, never on a verifier failure

→

OUTPUT · EXACTLY ONE OF

VERIFIED_REPORT

FALLBACK_SUMMARY REVIEW_ONLY

REJECTED

in `agentic/runtime/outbox/` ; anything not verified also goes to `runtime/review/`

`python -m agentic run-packet <packet.json>` `python -m agentic run-edf <night.edf>` `Path B dev: 31/31 verified`

All phases Rules that make the numbers trustworthy

Test set locked
29 nights, read once at the end, checksum-verified.

Validation fits everything
Every threshold, temperature and tier comes from the 31 dev nights.

Frozen gate
The verifier does not change while models are compared against it.

One shot per night
Temperature 0, no retries for quality, no best-of-N.

Rules written first
"≥ 25/31 at 5/5 → train" was fixed before any results existed.

Full text version with every file path: `PIPELINE_FLOW_README.md` at the repository root. Sources: packet `SC4081E0-PSG.json`, `README_FOR_CODE_REVIEW.md`, `ADDING EOG VERSION - FINAL SUMMARY/README.md`, `agentic/README.md`.